HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 04 Name: _____ Marks: / 26

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $a > b$ and $k < 0$, which of the following must be true?

   I. $a^2 > b^2$

   II. $a + k > b + k$

   III. $\frac{a}{k^2} > \frac{b}{k^2}$

   A. I only

   B. II only

   C. I and III only

   D. II and III only

(DSE-2014 PII Q06)

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2. If the angle and the radius of a sector are decreased by $x\%$ and 50% respectively so that its area is decreased by 90%, then $x =$

A. 20.

B. 40.

C. 60.

D. 80.

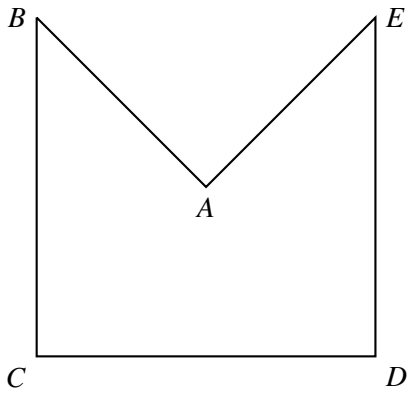
(DSE-2014 PII Q10)

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3. In the figure, $AB = AE$ and $\angle BAE = \angle BCD = \angle CDE = 90°$. If $BC = CD = DE = 16$ cm, then the area of the pentagon $ABCDE$ is




A. 71 $cm^2$.

B. 128 $cm^2$.

C. 192 $cm^2$.

D. 224 $cm^2$.

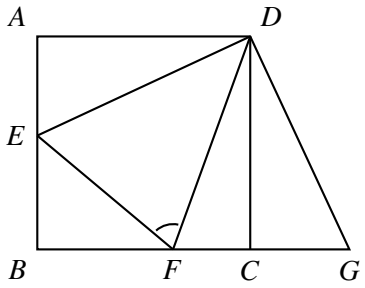
(DSE-2014 PII Q15)

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4. In the figure, $ABCD$ is a square. $BC$ is produced to $G$ such that $\angle CDG = 25°$. $E$ is a point lying on $AB$ such that $AE = CG$. If $F$ is a point lying on $BC$ such that $\angle CDF = 20°$, then $\angle DFE =$




A. 60°.

B. 65°.

C. 70°.

D. 73°.

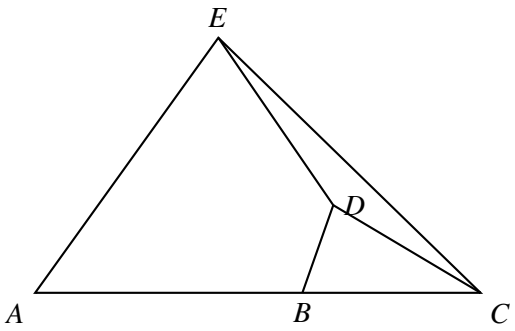
(DSE-2014 PII Q16)

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5. In the figure, $B$ is a point lying on $AC$ such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm$^2$ and 8 cm$^2$ respectively, then the area of the trapezium $ABDE$ is




A. 18 cm$^2$.

B. 21 cm$^2$.

C. 27 cm$^2$.

D. 33 cm$^2$.

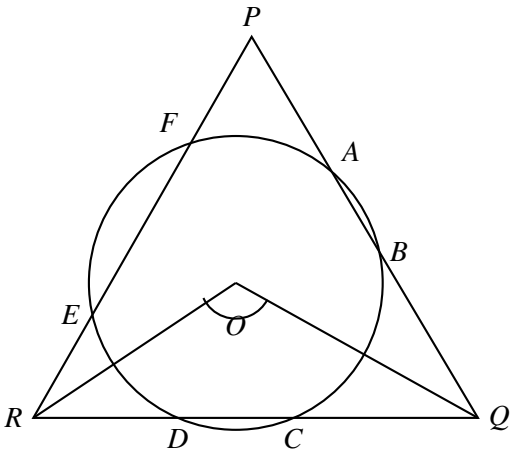
(DSE-2014 PII Q17)

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6. In the figure, $O$ is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at $A$, $B$, $C$, $D$, $E$ and $F$. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$




A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

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7. If a diameter of the circle $x^2 + y^2 - 8x + ky - 214 = 0$ passes through the point $(6, -5)$ and the slope of the diameter is $-4$, then $k =$

A. $-6$.

B. $-4$.

C. 13.

D. 70.

(DSE-2014 PII Q26)

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8. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

| Stem (tens) | Leaf (units) |
|---|---|
| 1 | $h$ 4 6 |
| 2 | 3 3 3 4 6 7 7 |
| 3 | 1 2 2 2 6 8 |
| 4 | 0 $k$ |

If the range of the above distribution is at least 33, which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

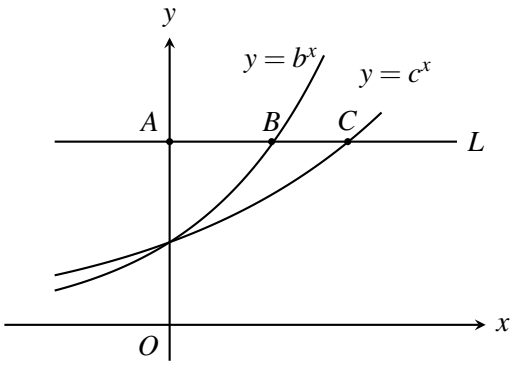
(DSE-2014 PII Q30)

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9. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where $b$ and $c$ are positive constants. If a horizontal line $L$ cuts the $y$-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at $A$, $B$ and $C$ respectively, which of the following are true?




I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

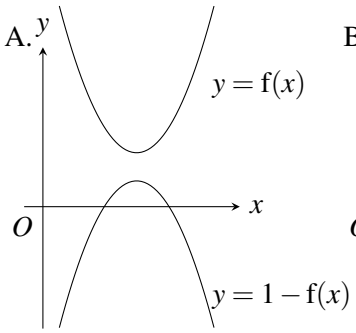
(DSE-2014 PII Q32)

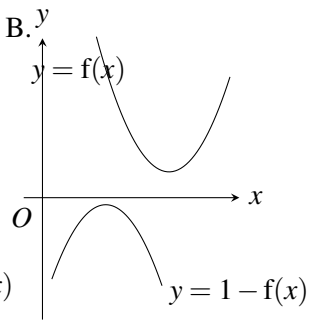
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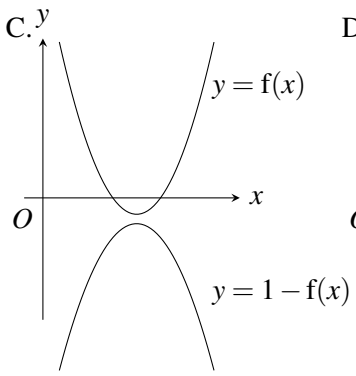
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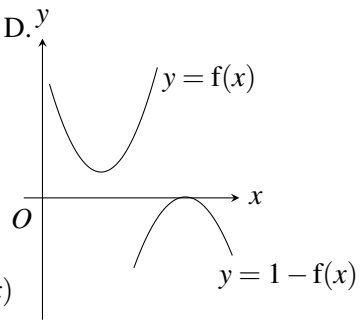
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10. Which of the following may represent the graph of $y = \mathrm{f}(x)$ and the graph of $y = 1 - \mathrm{f}(x)$ on the same rectangular coordinate system?

A.



B.



C.



D.



(DSE-2014 PII Q38)

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11. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

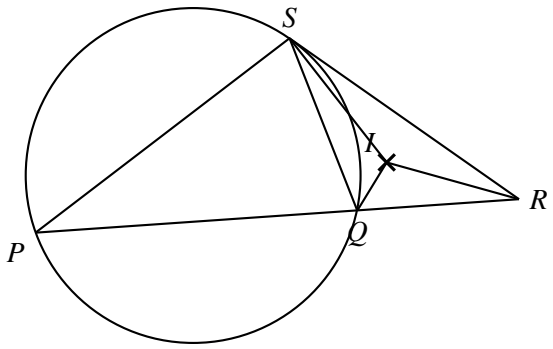
(DSE-2014 PII Q39)

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12. In the figure, $PQS$ is a circle. $PQ$ is produced to $R$ such that $RS$ is the tangent to the circle at $S$. $I$ is the in-centre of $\triangle QRS$. If $\angle IRQ = 12^\circ$ and $\angle PSQ = 70^\circ$, then $\angle QPS =$




A. $24^\circ$.

B. $37^\circ$.

C. $43^\circ$.

D. $62^\circ$.

(DSE-2014 PII Q41)

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13. If the straight line $x - y = k$ and the circle $x^2 + y^2 + 2x - 4y - 1 = 0$ intersect at $A$ and $B$, then the $x$-coordinate of the mid-point of $AB$ is

A. $1 + k$.

B. $1 - k$.

C. $\frac{1+k}{2}$.

D. $\frac{1-k}{2}$.

(DSE-2014 PII Q42)

I can finish _____ questions

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